Zachary Andrew Wilson

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Objective

To find co-op or internship opportunities in the field of Mechanical Engineering.

Availability
May – Aug 2027
Jan – Aug 2028

Education

- Rochester Institute of Technology, Mechanical Engineering: Energy and Environment BS — Expected May 2030
  - 3.97/4 Cumulative GPA
  - Dean's List: Fall 2025, Spring 2026
  - Classes of note: Circuits, Statics, Dynamics (current), Thermodynamics (current), Engineering Measurements Lab (current)
- Loudoun Academy of Engineering and Technology, Engineering Program, Summa Cum Laude — May 2025

Technical Skills & Certifications

- LinkedIn Learning Certification in Engineering Drawing for Manufacturing
- Trained in milling machine, lathe, drill press, assorted saws, FDM 3D printers, and CO_2 & diode laser cutters
- Adept at Autodesk Inventor, Onshape, Fusion 360, Google Workspace, and Microsoft Office Suites
- Competent in MATLAB, Ni LabVIEW, C++(Arduino IDE), and HTML

Projects

- *Automatic Chime Machine* — Jan. – Apr. 2026
  - Created design for and built a machine that automatically rotated and struck metal chimes to play the *Tetris* theme song, as a member of a team of four using Onshape, Microsoft Excel, and Arduino IDE
  - Personally developed frame to hold the chimes and engineered a solution to minimize potentiometer noise
  - Finished product successfully played the programmed 40 second song with the intended tempo across many consecutive runs
  - Presented at Imagine RIT 2026 Symposium, chosen to be kept as a future example by department
- *"Evaluating the Effectiveness of a Rack and Pinion Mechanism in Generating Power Using Variable Height Road Panels"* — Apr. 2024 – May 2025
  - Designed and built two-way linear to one-way rotary actuator to generate power from cars near stop signs using a depressing road panel on a team of two using Autodesk Inventor and Microsoft Excel
  - Devised the linear to rotational motion actuator, managed Inventor simulations, and headed testing
  - Successfully showed that the design was capable of power generation
  - Presented at the 2025 Academies of Loudoun Senior Symposium, received Project of Distinction Award
- *"Therapeutic Forearm and Wrist Orthosis for The Treatment of Carpal Tunnel Syndrome"* — Mar. - Apr 2024
  - Made an orthotic device to treat carpal tunnel that could apply heat and pressure to the wrist and forearm, as a member of a team of four using Autodesk Inventor and Arduino IDE
  - Designed and fabricated the orthosis sleeve and brace
  - Conducted and organized qualitative testing, subjects reported positive results for pain relief

Work Experience

- Domino's Pizza, Delivery Expert — May 2025 – Present
  - Work on a team in a fast-paced environment, ensuring punctual delivery of orders
- Glen Echo Park Aquarium, Volunteer/Part Time Education Assistant — Sep. 2015 – Present
  - Explain exhibits to guests, sell tickets, care for animals, designed website, and manage special projects
- DoorDash, Delivery Driver — Mar. – Apr. 2025